Many-body perturbation theory: diagram rules, third order and the linked-diagram theorem

FYS4480/FYS9480 – Week 45, November 5–6, 2026

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Thursday: resolvents, Wick contractions, symmetry factors and fermion signs
Friday: third-order diagrams, a proof of the linked-diagram theorem, and Pauli exclusion at fourth order

Reading: lecture notes chapter 9, section 9.15 (diagrammatic approach and the linked diagram theorem); Shavitt and Bartlett, chapters 4–6. Code: `BookPrograms/chapter09/diagrams.py`; notebook `week45.ipynb`.

Answers to the week-44 exercises and the week-45 exercises: last frames.

Plan for week 45

Thursday, 2×45 min: the lists, and the rules derived

- From week 44: the recursion, $\Omega^{(1,2,3)}$ and $E^{(1,2,3,4)}$ in resolvent form and in the determinant basis
- Normal ordering the vertices, $H_I = E^{(1)} + W$, absorbs every renormalisation term with a factor $E^{(1)}$: $E^{(3)} = \langle WR_0WR_0W\rangle_0$
- Contractions are lines; cuts are denominators ($[H_0, a_p^\dagger] = \varepsilon_p a_p^\dagger$); what Q removes
- The weight (4^{n_V} relabellings, equivalent lines) and the sign (vertex ends, the permutation π, $(-1)^{K-I}(-1)^p = (-1)^{h+l}$)
- First and second order re-read with the rules

Friday, 2×45 min: third order and the linked-diagram theorem

- Third order by intermediate states: fourteen connected diagrams, three of them MP3 (pp ladder, hh ladder, ph ring); the pairing model diagram by diagram, $-\frac{53}{576} - \frac{53}{576} + \frac{29}{288} = -\frac{1}{12}$ (units of g^3)
- Fourth order: the unlinked diagrams, the factorisation of their denominators, and the folded term $-E^{(2)}\langle WR_0^2W\rangle_0$ that removes them
- Proof of the linked-diagram theorem for the energy and wave operator. Exchange of repeated orbital labels at fourth order and the two-level benchmark

Where we are, and the route through this week

Done

- Full CI (weeks 37–38): exact, exponential; the amplitude equations drawn as diagrams with four rules we did not prove
- Hartree-Fock (weeks 38–39): $f_{ai} = 0$
- Week 44: P, Q, the resolvent, BW versus RS, $E^{(2)}$ from Wick's theorem, a two-level benchmark

This week

1. The complete lists through fourth order
2. The four rules, with proofs
3. Third order: fourteen diagrams
4. Fourth-order cancellation, the all-orders proof and the Pauli principle

Many-body connection

Notation as in week 44: $H = H_0 + H_I$ with $H_0 = \sum_p \varepsilon_p a_p^\dagger a_p$, reference $|\Phi_0\rangle$ with $\varepsilon_0 = \sum_i \varepsilon_i$, $R_0 = Q/(\varepsilon_0 - H_0)$, $D_m = \varepsilon_0 - \varepsilon_m$, $\langle X\rangle_0 = \langle\Phi_0|X|\Phi_0\rangle$, $V_{mn} = \langle\Phi_m|H_I|\Phi_n\rangle$. Occupied i, j, k, l, empty a, b, c, d. Antisymmetrised elements $\overline{v}_{pqrs} = \langle pq|v|rs\rangle_{\mathrm{AS}}$ throughout.

Thursday, lecture 1: The lists, and from sums to pictures

The recursion that generates every order (week 44, lecture 4)

With $R_0 = Q/(\varepsilon_0 - H_0)$ and $|\Psi_0\rangle = \sum_n |\Psi^{(n)}\rangle$, $|\Psi^{(0)}\rangle = |\Phi_0\rangle$, collecting powers of η in $|\Psi_0\rangle = |\Phi_0\rangle + R_0(H_I - \Delta E)|\Psi_0\rangle$ gives

$$\boxed{|\Psi^{(n)}\rangle = R_0\Big[H_I|\Psi^{(n-1)}\rangle - \sum_{k=1}^{n-1} E^{(k)}|\Psi^{(n-k)}\rangle\Big], \qquad E^{(n+1)} = \langle\Phi_0|H_I|\Psi^{(n)}\rangle.}$$

- The subtracted terms are the normalisation ("renormalisation", "unlinked") corrections; they appear from $n = 2$ on.
- $R_0|\Phi_0\rangle = 0$: every $|\Psi^{(n)}\rangle$, $n \geq 1$, lies in Q space, so intermediate normalisation is preserved order by order.
- `mbpt.py` implements exactly this recursion and checks it against the explicit expressions on the next two frames.

All contributions to the wave operator through third order

Write $|\Psi_0\rangle = \Omega|\Phi_0\rangle$ with $\Omega = 1 + \Omega^{(1)} + \Omega^{(2)} + \Omega^{(3)} + \cdots$, $R \equiv R_0$, $V \equiv H_I$:

$$\Omega^{(1)} = RV,$$

$$\Omega^{(2)} = RVRV \; - \; E^{(1)} R^2 V,$$

$$\Omega^{(3)} = RVRVRV \; - \; E^{(1)}\left[RVR^2V + R^2VRV\right] \; + \; \left(E^{(1)}\right)^2 R^3 V \; - \; E^{(2)} R^2 V.$$

In the determinant basis, with $V_{mn} = \langle\Phi_m|H_I|\Phi_n\rangle$ and $D_m = \varepsilon_0 - \varepsilon_m$ $(m, n, I \neq 0)$:

$$\langle\Phi_m|\Psi^{(1)}\rangle = \frac{V_{m0}}{D_m}, \qquad \langle\Phi_m|\Psi^{(2)}\rangle = \sum_n \frac{V_{mn}V_{n0}}{D_m D_n} - \frac{V_{00}V_{m0}}{D_m^2},$$

$$\langle\Phi_m|\Psi^{(3)}\rangle = \sum_{nI} \frac{V_{mn}V_{nI}V_{I0}}{D_m D_n D_I} - V_{00}\Big[\sum_n \frac{V_{mn}V_{n0}}{D_m D_n^2} + \sum_n \frac{V_{mn}V_{n0}}{D_m^2 D_n}\Big] + V_{00}^2 \frac{V_{m0}}{D_m^3} - E^{(2)}\frac{V_{m0}}{D_m^2}.$$

For a two-body H_I: $\Psi^{(1)}$ has singles ($U_{ai}/(\varepsilon_i - \varepsilon_a)$) and doubles ($\overline{v}_{abij}/(\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b)$); $\Psi^{(2)}$ reaches up to quadruples; $\Psi^{(3)}$ up to sextuples. Rank is not order.

All contributions to the energy through fourth order

With $\langle X\rangle_0 = \langle\Phi_0|X|\Phi_0\rangle$, $R = Q/(\varepsilon_0 - H_0)$, $V = H_I$, and $E^{(n+1)} = \langle V\Omega^{(n)}\rangle_0$:

$$E^{(1)} = \langle V\rangle_0,$$

$$E^{(2)} = \langle VRV\rangle_0,$$

$$E^{(3)} = \langle VRVRV\rangle_0 \; - \; E^{(1)}\langle VR^2V\rangle_0,$$

$$E^{(4)} = \langle VRVRVRV\rangle_0 \; - \; E^{(1)}\left[\langle VR^2VRV\rangle_0 + \langle VRVR^2V\rangle_0\right] \; + \; \left(E^{(1)}\right)^2\langle VR^3V\rangle_0 \; - \; E^{(2)}\langle VR^2V\rangle_0.$$

In the determinant basis:

$$E^{(1)} = V_{00}, \quad E^{(2)} = \sum_m \frac{V_{0m}V_{m0}}{D_m}, \quad E^{(3)} = \sum_{mn}\frac{V_{0m}V_{mn}V_{n0}}{D_mD_n} - V_{00}\sum_m\frac{V_{0m}V_{m0}}{D_m^2},$$

$$E^{(4)} = \sum_{mnl}\frac{V_{0m}V_{mn}V_{nl}V_{l0}}{D_mD_nD_l} - 2V_{00}\sum_{mn}\frac{V_{0m}V_{mn}V_{n0}}{D_m^2D_n} + V_{00}^2\sum_m\frac{V_{0m}V_{m0}}{D_m^3} - E^{(2)}\sum_m\frac{V_{0m}V_{m0}}{D_m^2}.$$

(The two mixed terms are complex conjugates, equal for real matrix elements, hence the 2.) Every term with a factor $E^{(k)}$ in front is a renormalisation term: the product of two separately evaluated pieces. Today: these are the *unlinked* diagrams, and they cancel against the unlinked parts of the principal term – the linked-diagram theorem.

The same list in the language of determinants, through second order

For $H_0 = \sum_p \varepsilon_p a_p^\dagger a_p$ and $H_I = E^{(1)} + U_N + V_N$:

$$E^{(1)} = \sum_i u_{ii} + \frac{1}{2} \sum_{ij \in \mathrm{occ}} \overline{v}_{ijij} \quad \text{(with } u = 0 \text{ for a diagonal } h_0\text{)},$$

$$E^{(2)} = \underbrace{\sum_{ia} \frac{|U_{ai}|^2}{\varepsilon_i - \varepsilon_a}}_{1p1h} + \underbrace{\frac{1}{4} \sum_{ijab} \frac{|\overline{v}_{abij}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}}_{2p2h},$$

$$\Omega^{(1)}|\Phi_0\rangle = \sum_{ia} \frac{U_{ai}}{\varepsilon_i - \varepsilon_a} |\Phi_i^a\rangle + \frac{1}{4} \sum_{ijab} \frac{\overline{v}_{abij}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b} |\Phi_{ij}^{ab}\rangle.$$

Measured

Four models, `mbpt.py`: the fourth order from the explicit formula and from the recursion agree to 10^{-12}. Hubbard ($U = 2$): the four pieces are -0.191, $+0.303$, -0.151, $+0.032$ and the total -0.0076. Pairing ($g = 0.5$): $E^{(3)}$ is $-0.02300347 - (-0.01258681) = -0.01041667$. The cancellations are severe, and they are the point of this week.

Pause to think #1

(about 3 min – discuss with your neighbour)

Look at the list of fourth-order terms.

1. How many distinct terms does $E^{(4)}$ have for a reference with $E^{(1)} = 0$ (Hartree-Fock partition with the mean field in H_0)?
2. In the third order, $E^{(1)}\langle VR^2V\rangle_0$ is the product of two quantities that are each sums over the whole system. If the system were two non-interacting copies of a smaller one, how would this product scale with the number of copies? And $E^{(3)}$ itself?

Answer to pause #1

1. Two: the principal term $\langle VRVRVRV\rangle_0$ and the folded term $-E^{(2)}\langle VR^2V\rangle_0$. For the Lipkin model ($E^{(1)}=0$) `mbpt.py` gives -0.0036 and $+0.0072$, total $+0.0036$.
2. $E^{(1)}$ and $\langle VR^2V\rangle_0$ each scale linearly with the number of copies, so their product scales *quadratically* – an energy cannot. The principal term $\langle VRVRV\rangle_0$ contains a quadratically scaling piece of its own, and the subtraction removes it exactly: $E^{(3)}$ scales linearly. This is size extensivity, verified to machine precision in chapter 9, and on Friday it becomes the statement that only *linked* diagrams contribute.

Step zero: normal ordering absorbs the $E^{(1)}$ renormalisation terms

Write the perturbation in the normal-ordered form of week 44 (section 9.8), with a name for its non-scalar part:

$$H_I = E^{(1)} + W, \qquad \boxed{W = \sum_{pq} \tilde{f}_{pq} \{ a_p^\dagger a_q \} + \frac{1}{4} \sum_{pqrs} \overline{v}_{pqrs} \{ a_p^\dagger a_q^\dagger a_s a_r \},} \qquad \tilde{f}_{pq} = u_{pq} + U_{pq} = f_{pq} - \varepsilon_p \delta_{pq},$$

where f is the Fock matrix of week 39 and $U_{pq} = \sum_j \overline{v}_{pjqj}$ the mean field of the reference. $\langle W \rangle_0 = 0$, and every normal-ordered product has a vanishing reference expectation value.

Look at the recursion

The $k = 1$ term subtracts $E^{(1)} |\Psi^{(n-1)}\rangle$ from $H_I |\Psi^{(n-1)}\rangle$; the two combine into $W |\Psi^{(n-1)}\rangle$. Since $\langle \Phi_0 | \Psi^{(n)} \rangle = 0$ for $n \geq 1$, the energy formula may use W too:

$$\boxed{|\Psi^{(n)}\rangle = R_0 \Big[W |\Psi^{(n-1)}\rangle - \sum_{k=2}^{n-1} E^{(k)} |\Psi^{(n-k)}\rangle \Big], \quad E^{(n+1)} = \langle \Phi_0 | W | \Psi^{(n)} \rangle \ (n \geq 1).}$$

$E^{(1)}$ has disappeared from the recursion, and with it *every* renormalisation term carrying a factor $E^{(1)}$.

The orders with normal-ordered vertices

$$E^{(2)} = \langle WR_0W\rangle_0, \qquad E^{(3)} = \langle WR_0WR_0W\rangle_0, \qquad E^{(4)} = \langle WR_0WR_0WR_0W\rangle_0 - E^{(2)}\langle WR_0^2W\rangle_0.$$

Third order: *nothing to subtract*. Fourth order: only the folded term is left.

The identity behind it is one line

In the determinant basis $W_{mn} = V_{mn} - V_{00}\delta_{mn}$, so

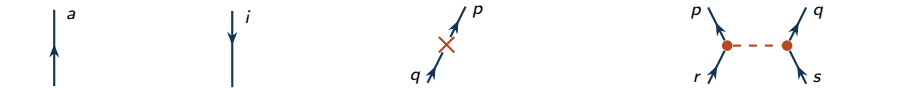
$$\sum_{m,n\neq 0} \frac{V_{0m}W_{mn}V_{n0}}{D_mD_n} = \sum_{m,n\neq 0} \frac{V_{0m}V_{mn}V_{n0}}{D_mD_n} - V_{00}\sum_{m\neq 0}\frac{V_{0m}V_{m0}}{D_m^2}\,:$$

the renormalisation term of third order *is* the scalar part of the middle vertex, removed by not putting it there.

Measured

Pairing model, $g = 0.5$: $\langle WR_0WR_0W\rangle_0 = -0.01041667$, the whole third-order energy in one term. From here on every vertex is normal ordered: a cross carries $\tilde{f}_{pq}$ (cross plus bubble of week 44), an interaction the antisymmetrised element.

The elements of a diagram (week 38, chapter 5)



particle line | hole line | $\tilde{f}_{pq}$ (out p, in q) | $\overline{v}_{pqrs}$: ends (p,r) and (q,s)

- Time runs upward; the reference $|\Phi_0\rangle$ is the vacuum and is not drawn. Particle line: arrow up; hole line: arrow down.
- Every vertex is a product of *ends*, an end being $a^\dagger_{\rm out}\, a_{\rm in}$: the cross is one end, the interaction two ends joined by the dashed line, with the pairing (p,r), (q,s) of the matrix element.
- New this week: between successive vertices sits R_0; a horizontal *cut* between them will give the denominator.

The four rules of week 38 (sums, denominators, $\frac{1}{2}$ per pair of equivalent lines, $(-1)^{h+l}$) were stated and checked, not proved. Today they are proved.

A term of order n is a string, read from the bottom

$$\langle WR_0WR_0\cdots R_0W\rangle_0 = \langle\Phi_0|W_nR_0W_{n-1}R_0\cdots R_0W_1|\Phi_0\rangle,$$

the subscript numbering the vertices from the one that acts first (rightmost, bottom) to the one that acts last (leftmost, top).

Forget the resolvents for a moment

Each W_k is a sum of normal-ordered monomials. The generalised Wick theorem (week 37, section 3.13): the vacuum expectation value of a product of normal-ordered groups is the sum over all *complete* contractions *between different groups*. Only two contractions are non-zero relative to $|\Phi_0\rangle$:

$$\boxed{\langle\Phi_0|a_a a_a^\dagger|\Phi_0\rangle = 1\ (a > F), \qquad \langle\Phi_0|a_i^\dagger a_i|\Phi_0\rangle = 1\ (i \le F),}$$

and in both the operator on the *left* belongs to the *later* vertex.

- First: $a_a^\dagger$ earlier creates a particle, a_a later absorbs it – a particle line, arrow up. Second: a_i earlier empties an occupied orbital, $a_i^\dagger$ later fills it – a hole line, arrow down.
- A complete contraction = a drawing in which every operator of every vertex is the end of exactly one line to a *different* vertex. Sum over contractions = sum over drawings: rule 1.

Pause to think #2 (about 3 min – discuss with your neighbour)

With normal-ordered vertices no line can begin and end at the same vertex.

1. What happened to the *bubbles* of last week's second-order diagrams (b), (c), (d) – a hole line leaving and re-entering the same interaction?
2. Draw the second-order diagrams that remain with normal-ordered vertices. How many are there?

Answer to pause #2

1. A bubble is a contraction *within* one vertex, $\sum_j \overline{v}_{pjqj} = U_{pq}$. Normal ordering has already carried it out: it is the U_{pq} inside $\tilde{f}_{pq} = u_{pq} + U_{pq}$. The cross now stands for "cross plus bubble", and the diagrams (b), (c), (d) of week 44 are the expansion of a single drawing in which the two crosses carry $\tilde{f}$.
2. Two: two crosses joined by a particle line and a hole line, and the ring – the interaction creating two pairs and the interaction annihilating them.

$$E^{(2)} = \sum_{ia} \frac{\tilde{f}_{ia}\tilde{f}_{ai}}{\varepsilon_i - \varepsilon_a} + \frac{1}{4}\sum_{ijab} \frac{|\overline{v}_{abij}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b},$$

and $\tilde{f}_{ai} = f_{ai}$ for $a \neq i$: the coherent singles sum of week 44 is one diagram.

Thursday, lecture 2: The four rules, proved

One Wick contraction gives one orbital sum

Write $n_p = 1$ for an occupied reference orbital and $n_p = 0$ for an empty one. The determinant contractions are

$$\langle a_p^\dagger a_q \rangle_0 = \delta_{pq} n_p, \qquad \langle a_q a_p^\dagger \rangle_0 = \delta_{pq}(1 - n_p).$$

For example, take one one-body vertex $F = \sum_{pq} \tilde{f}_{pq} \{ a_p^\dagger a_q \}$ in each of two successive positions in the operator string. Normal ordering leaves only the intervertex pairing

$$\langle \{ a_p^\dagger a_q \} \{ a_r^\dagger a_s \} \rangle_0 = \delta_{ps} n_p \, \delta_{qr} (1 - n_q).$$

The two Kronecker deltas identify the endpoints. The occupation factors restrict $p = s = i$ to holes and $q = r = a$ to particles. Thus $\langle FF \rangle_0 = \sum_{ia} \tilde{f}_{ia} \tilde{f}_{ai}$. Each internal line has one independently summed orbital label.

The reduced resolvent on a determinant

For a nondegenerate reference, let $Q = 1 - |\Phi_0\rangle\langle\Phi_0|$. Define the inverse on the range of Q:

$$R_0 = \sum_{m\neq 0} \frac{|\Phi_m\rangle\langle\Phi_m|}{\varepsilon_0 - \varepsilon_m}, \qquad R_0|\Phi_0\rangle = 0.$$

If M is a normal-ordered monomial, $[H_0, M] = \omega_M M$, where ω_M is the sum of the created orbital energies minus the sum of the annihilated ones. By the product rule for commutators,

$$[H_0, M_k \cdots M_1] = \Big(\sum_{j=1}^{k} \omega_{M_j}\Big) M_k \cdots M_1.$$

Whenever $M_k \cdots M_1|\Phi_0\rangle$ is nonzero, it has unperturbed energy $\varepsilon_0 + \sum_j \omega_{M_j}$. If a repeated occupied or empty label makes this state vanish, the determinant path already contributes zero.

Reading the energy denominator at a cut

A contraction entirely below a horizontal cut contributes $+\varepsilon_p - \varepsilon_p = 0$. An upward particle line leaves a creation operator below the cut. A downward hole line leaves an annihilation operator below it. Hence

$$D_k = \varepsilon_0 - \varepsilon_{m_k} = \sum_{i \text{ crossing}} \varepsilon_i - \sum_{a \text{ crossing}} \varepsilon_a.$$

For $|\Phi_{ij}^{ab}\rangle = a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle$ this gives

$$R_0 |\Phi_{ij}^{ab}\rangle = \frac{|\Phi_{ij}^{ab}\rangle}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}.$$

A closed, time-ordered term with n vertices has $n-1$ cuts. Multiply by $1/D_k$ at each cut. A cut with no crossing lines returns to the reference and Q kills the term. Two independent excitations can overlap in time and survive this test.

Counting contractions for antisymmetrised vertices

Each two-body vertex carries $1/(2!\,2!)$ and has two outgoing and two incoming slots. Exchanging two slots on one side reverses both the Wick sign and the sign of $\overline{v}_{pqrs}$, leaving their product unchanged.

For a fixed ordering of the vertices

At n_V two-body vertices there are 4^{n_V} choices of slot labels. Two choices give the same pairing when they exchange parallel lines at *both* endpoints. With at most two incoming and two outgoing slots, each pair of equivalent internal lines supplies two such choices.

$$\#\{\text{distinct pairings}\} = \frac{4^{n_V}}{2^{n_{\mathrm{eq}}}}, \qquad w = \frac{1}{4^{n_V}}\frac{4^{n_V}}{2^{n_{\mathrm{eq}}}} = 2^{-n_{\mathrm{eq}}}.$$

Equivalent lines have the same two vertices and the same arrow direction. Throughout this count the orbital sums are unrestricted within the particle and hole spaces. Exchanging externally fixed labels gives a different amplitude ordering.

Four contractions at second order

Take the top vertex as $\overline{v}_{pqrs}$ and the bottom vertex as $\overline{v}_{abij}$, with i, j occupied and a, b empty. The two particle contractions and the two hole contractions give

$$(\delta_{ra}\delta_{sb} - \delta_{rb}\delta_{sa})(\delta_{pi}\delta_{qj} - \delta_{pj}\delta_{qi}).$$

Contracting the top matrix element with both brackets gives $4\overline{v}_{ijab}$. Thus the two vertex factors and four Wick pairings give $(1/4)^2\, 4 = 1/4$.

$$E^{(2)}_{\text{doubles}} = \frac{1}{4}\sum_{ijab}\frac{|\overline{v}_{abij}|^2}{D^{ab}_{ij}} = \sum_{i<j}\sum_{a<b}\frac{|\overline{v}_{abij}|^2}{D^{ab}_{ij}}, \qquad D^{ab}_{ij} = \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b.$$

Every doubly excited determinant occurs once in the ordered sums. This checks the diagram weight independently against the determinant expression.

The sign, 1: vertex ends and the permutation π

The term of a complete contraction carries the sign of the permutation that brings the contracted pairs next to each other, in the order in which they appear (section 3.13). Reorder inside each vertex first:

$$\{a_p^\dagger a_q^\dagger a_s a_r\} = \{(a_p^\dagger a_r)(a_q^\dagger a_s)\}$$

(moving a_r two places left costs two signs; no contraction terms inside a normal-ordered product). So every vertex is a product of *ends* $a_{\mathrm{out}}^\dagger a_{\mathrm{in}}$, a cross being a single end. Number the ends $1, \ldots, K$ along the string.

Lines join an "out" to an "in"

A particle line contracts an $a_a^\dagger$ with an a_a, a hole line an a_i with a $a_i^\dagger$: in either case one operator is an out and the other an in. The lines therefore define a permutation π of the ends: the out-operator of end k is contracted with the in-operator of end $\pi(k)$.

- Follow a line to the end where it arrives, continue with the line leaving that end, ... until you are back: a closed loop is a cycle of π. So l = number of cycles.
- The number of lines equals the number of ends: $K = h + p$ (hole lines plus particle lines).

The sign, 2: $(-1)^{K-l}(-1)^{p} = (-1)^{h+l}$

Step 1: bring every pair into the order (out, in)

Place each $a^{\dagger}_{\mathrm{out}}$ immediately before the a_{in} it is contracted with. This permutes only the incoming operators among the incoming slots, by π; a transposition of two annihilation operators costs one sign whatever stands between them. Cost: the parity of π, $(-1)^{K-l}$ (a permutation of K objects with l cycles is a product of $K-l$ transpositions).

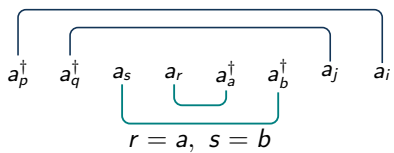
Step 2: compare with the order Wick's theorem wants (first in the string = first in the pair)

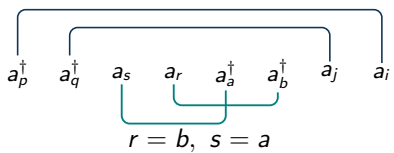
Hole line: $a_i^{\dagger}$ belongs to the later vertex and already stands left: value $\langle \Phi_0 | a_i^{\dagger} a_i | \Phi_0 \rangle = 1$, no extra sign.
Particle line: a_a belongs to the later vertex and should stand left: one more transposition, then $\langle \Phi_0 | a_a a_a^{\dagger} | \Phi_0 \rangle = 1$. Cost $(-1)^{p}$.

$$\boxed{(-1)^{K-l}(-1)^{p} = (-1)^{h+p-l+p} = (-1)^{h+l}} \qquad \text{rule 4.}$$

Exchanging two particle contractions changes one loop

Hold the hole contractions fixed in the second-order doubles diagram. In each string the four operators on the left belong to the upper vertex. Blue brackets fix $p = i$, $q = j$.





The exchange joins two cycles into one. At the same time the upper vertex changes from $\overline{v}_{ijab}$ to $\overline{v}_{ijba} = -\overline{v}_{ijab}$. The two signed terms therefore add. Exchanging the holes supplies the other two contractions counted above.

Open diagrams and the ordering of external lines

Fix $|\Phi_{ij}^{ab}\rangle = a_a^\dagger a_b^\dagger a_j a_i |\Phi_0\rangle$. To assign a sign to its amplitude, append the bra at the top of the diagram. Its operator string can be written

$$\langle \Phi_{ij}^{ab}| = \langle \Phi_0 | a_i^\dagger a_j^\dagger a_b a_a = \langle \Phi_0 | (a_i^\dagger a_a)(a_j^\dagger a_b).$$

Since a, b are virtual and i, j are occupied, moving a_a through the two middle operators produces two minus signs and no delta term.

- Pair (a, i) and (b, j) at the bra. Count all hole lines and cycles of this *closed* contraction, including those running to the bra. A loop closed this way is called a quasi-loop.
- Exchanging a and b in the external ordering reverses the determinant sign. When summing inequivalent attachments use the antisymmetriser $1 - P_{ab}$, where P_{ab} simply interchanges labels. The same convention applies to i, j.

The projecting bra carries unit weight. A wave-function amplitude with n physical vertices has an additional cut between the last vertex and the bra, for n denominators in total.

Pause to think #3

(about 4 min – discuss with your neighbour)

Apply the two proofs to the second-order ring (interaction creating two pairs, interaction annihilating them) and to the two-cross diagram.

1. List the ends and the permutation π for each. How many cycles? How many hole lines? What sign?
2. How many pairs of equivalent lines does each have, and what is the weight?
3. Which lines cross the single cut, and what is the denominator?

Answer to pause #3

1. Ring: ends $1L = (a, i)$, $1R = (b, j)$ (bottom), $2L = (i, a)$, $2R = (j, b)$ (top), writing (out, in). Lines: $1L \to 2L$ (a), $2L \to 1L$ (i): one cycle; $1R \to 2R$, $2R \to 1R$: a second. $l = 2$, $h = 2$: sign $+$. Two crosses: ends (a, i), (i, a), one 2-cycle, $l = 1$, $h = 1$: sign $+$.
2. Ring: the particle lines a, b both run from vertex 1 to vertex 2, the hole lines i, j both from 2 to 1 – two pairs, weight $\frac{1}{4}$. Two crosses: none, weight 1.
3. Ring: four lines cross, $1/(\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b)$. Two crosses: $1/(\varepsilon_i - \varepsilon_a)$. Together: $E^{(2)}$ of pause #2, now *derived* rather than checked.

The four rules for a closed, time-ordered diagram

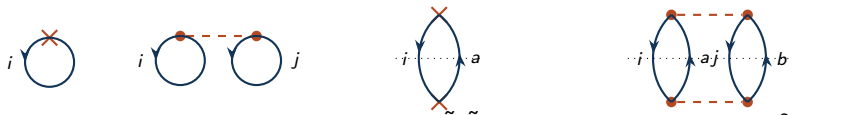
Use normal-ordered vertices, antisymmetrised two-body matrix elements and unrestricted internal orbital sums. For each distinct time ordering:

1. Assign one label per internal line. Sum holes over occupied orbitals and particles over empty orbitals. At every cross write $\tilde{f}_{pq}$, and at every interaction write $\overline{v}_{pqrs}$ with ends (p, r) and (q, s).
2. Insert $1/D_k$ between consecutive vertices, with $D_k = \sum_{\text{holes}} \varepsilon_i - \sum_{\text{particles}} \varepsilon_a$ on that cut.
3. Multiply by $2^{-n_{\text{eq}}}$ for the equivalent internal line pairs.
4. Multiply by $(-1)^{h+l}$ using the hole lines and closed fermion loops of the chosen vertex pairing.

Sign check

The Wick sign and the signs of the denominators are separate. The doubles diagram has $h = l = 2$, so its Wick sign is $+1$. For a ground-state reference $D_{ij}^{ab} < 0$, and its contribution to $E^{(2)}$ is negative whenever the matrix element is nonzero.

First and second order, re-read with the rules



$\sum_i u_{ii}$ $\quad \frac{1}{2}\sum_{ij} \overline{v}_{ijij}$ $\quad \sum_{ia} \frac{\tilde{f}_{ia}\tilde{f}_{ai}}{\varepsilon_i - \varepsilon_a}$ $\quad \frac{1}{4}\sum_{ijab} \frac{|\overline{v}_{abij}|^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$

- First order uses the bare vertices. The two-body expectation is $\frac{1}{2}\sum_{ij} \overline{v}_{ijij}$ from two Wick pairings times $1/4$. Both diagrams have positive Wick sign and no cut. Together they give $E_{\mathrm{ref}} - \varepsilon_0$.
- Second order (normal-ordered vertices): the two diagrams of pause #3.
- Hartree-Fock basis: f is diagonal with eigenvalues ε_p, so $\tilde{f}_{pq} = 0$ for all p, q: every cross vanishes, every bubble with it (Brillouin), and only the ring survives – MP2. Pairing model: $\tilde{f}_{ai} = 0$ in the pairing basis, ring $= -\frac{7}{24}g^2$.

Friday, lecture 3: Third order

Enumerating the third order by intermediate states

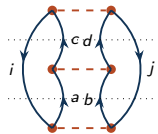
$E^{(3)} = \langle WR_0WR_0W\rangle_0$: all connected drawings with three normal-ordered vertices. The intermediate states can only be singles (S) or doubles (D):

- Bottom vertex on the reference: a cross creates a single ($\tilde{f}_{ai}$), an interaction a double ($\overline{v}_{abij}$) – a normal-ordered operator has to *create* every line it touches.
- Top vertex: a cross closes a single ($\tilde{f}_{ia}$), an interaction a double ($\overline{v}_{ijab}$).
- Middle vertex: S→S, S→D, D→S, D→D, by a cross or by an interaction. Triples and quadruples cannot be closed by the top vertex; a return to the reference is killed by Q.

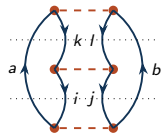
	vertices (bottom to top)	states	description	sign, weight
1, 2	cross, cross, cross	S, S	particle-line insertion / hole-line insertion	$+$ / $-$
3	cross, cross, interaction	S, D	two pairs created by crosses	$+$
4	cross, interaction, cross	S, S	particle-hole scattering	$-$
5, 6	cross, interaction, interaction	S, D	particle line / hole line into the interaction	$+\frac{1}{2}$ / $-\frac{1}{2}$
7, 8	interaction, cross, interaction	D, D	particle-line / hole-line insertion	$+\frac{1}{2}$ / $-\frac{1}{2}$
9	interaction, cross, cross	D, S	two pairs closed by crosses	$+$
10, 11	interaction, interaction, interaction	D, D	particle-particle ladder / hole-hole ladder	$+\frac{1}{8}$ / $+\frac{1}{8}$
12	interaction, interaction, interaction	D, D	particle-hole ring	$-$
13, 14	interaction, interaction, cross	D, S	particle lines / hole lines into the interaction	$+\frac{1}{2}$ / $-\frac{1}{2}$

Fourteen diagrams. Nothing forgotten, nothing counted twice: that is what the pictures are for.

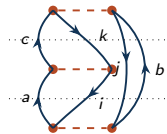
The three diagrams with three interactions: MP3



(10)



(11)



(12)

$$\Delta E^{(3)}_{\mathrm{MP}} = \frac{1}{8}\sum \frac{\overline{v}_{ijcd}\overline{v}_{cdab}\overline{v}_{abij}}{\varepsilon^{ab}_{ij}\,\varepsilon^{cd}_{ij}} + \frac{1}{8}\sum \frac{\overline{v}_{klab}\overline{v}_{ijkl}\overline{v}_{abij}}{\varepsilon^{ab}_{ij}\,\varepsilon^{ab}_{kl}} + \sum \frac{\overline{v}_{kjcb}\overline{v}_{icak}\overline{v}_{abij}}{\varepsilon^{ab}_{ij}\,\varepsilon^{cb}_{kj}},$$

with $\varepsilon^{ab}_{ij} \equiv \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b$.

Ladders: three pairs of equivalent lines, $\frac{1}{8}$; $h = 2$ or 4, $l = 2$: $+$. Ring: no equivalent lines, $h = 3$, $l = 2$: $-$ in front of $\overline{v}_{ciak}$, i.e. $+$ with $\overline{v}_{icak}$.

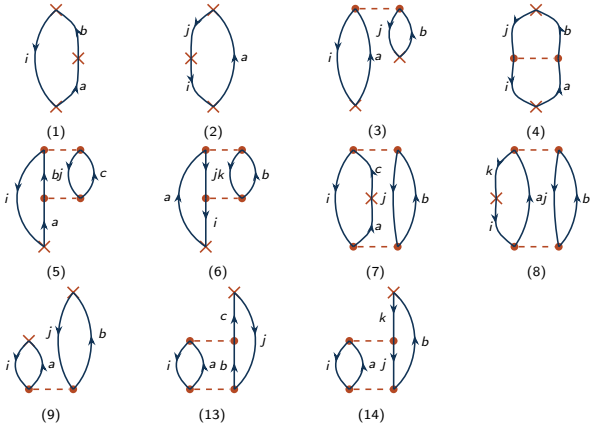
MP3: what the three diagrams say

- In a canonical Hartree-Fock basis $\tilde{f}_{pq} = f_{pq} - \varepsilon_p \delta_{pq} = 0$: every diagram with a cross vanishes, and the three are the *only* third-order diagrams – the Møller-Plesset third-order energy.
- Particle-particle ladder: the two excited particles scatter off each other once before they fall back. Hole-hole ladder: the same for the two holes. Ring: one particle-hole pair scatters into another.
- Each is the third-order member of an infinite family. Ladders of every length: Brueckner's G-matrix of nuclear physics. Rings of every length: the RPA of chapter 7 (week 40, and the stability matrix of week 39). Both at once, with the terms that mix them: coupled-cluster doubles, chapter 10.

Many-body connection

Why one may sum a family to all orders and still get a meaningful answer is the content of the linked-diagram theorem (lecture 4): the pieces being summed are the ones that count, and no unlinked piece sneaks in.

The eleven diagrams with at least one cross



All vanish in a canonical HF basis. In the models of chapter 9 (H_0 diagonal by construction, reference not HF) the cross is the bubble U_{pq}: 7 and 8 are the first self-energy insertions on a particle and a hole line. Mirror images in time: 3/9, 5/13, 6/14.

The full third order, read off the drawings: diagrams 1–7

$$
\begin{aligned}
E^{(3)} = &\sum_{iab} \frac{\tilde{f}_{ib}\tilde{f}_{ba}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)(\varepsilon_i-\varepsilon_b)} - \sum_{ija} \frac{\tilde{f}_{ja}\tilde{f}_{ij}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)(\varepsilon_j-\varepsilon_a)} + \sum_{ijab} \frac{\overline{v}_{ijab}\tilde{f}_{bj}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)\,\varepsilon_{ij}^{ab}} \\
&- \sum_{ijab} \frac{\tilde{f}_{jb}\overline{v}_{ibja}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)(\varepsilon_j-\varepsilon_b)} + \frac{1}{2}\sum \frac{\overline{v}_{ijbc}\overline{v}_{bcaj}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)\,\varepsilon_{ij}^{bc}} - \frac{1}{2}\sum \frac{\overline{v}_{jkab}\overline{v}_{ibjk}\tilde{f}_{ai}}{(\varepsilon_i-\varepsilon_a)\,\varepsilon_{jk}^{ab}} \\
&+ \frac{1}{2}\sum \frac{\overline{v}_{ijcb}\tilde{f}_{ca}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\,\varepsilon_{ij}^{cb}} + \cdots \qquad (\varepsilon_{ij}^{ab} = \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b)
\end{aligned}
$$

A sign worth checking: diagram 2, the cross on a hole line

Ends $(a,i),(i,j),(j,a)$: π is one 3-cycle, $l=1$; $h=2$: sign $-$. The cross adds a line, not a loop; whether the added line is a hole or a particle decides the sign – the $f_{ab}C_i^b$ versus $-f_{ji}C_j^a$ of the singles equation of week 38. Diagram 1 (cross on the particle line) has the same single cycle but $h=1$: $+$.

The full third order, read off the drawings: diagrams 8–14

$$\cdots - \frac{1}{2}\sum \frac{\overline{v}_{kjab}\tilde{f}_{ik}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\varepsilon_{kj}^{ab}} + \sum_{ijab} \frac{\tilde{f}_{jb}\tilde{f}_{ia}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}(\varepsilon_j - \varepsilon_b)} + \frac{1}{8}\sum \frac{\overline{v}_{ijcd}\overline{v}_{cdab}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\varepsilon_{ij}^{cd}}$$
$$+ \frac{1}{8}\sum \frac{\overline{v}_{klab}\overline{v}_{ijkl}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\varepsilon_{kl}^{ab}} - \sum \frac{\overline{v}_{kjcb}\overline{v}_{ciak}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}\varepsilon_{kj}^{cb}}$$
$$+ \frac{1}{2}\sum \frac{\tilde{f}_{jc}\overline{v}_{icab}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}(\varepsilon_j - \varepsilon_c)} - \frac{1}{2}\sum \frac{\tilde{f}_{kb}\overline{v}_{ijak}\overline{v}_{abij}}{\varepsilon_{ij}^{ab}(\varepsilon_k - \varepsilon_b)}.$$

A sign worth checking: diagram 12, the ring

Ends $1L{=}(a,i)$, $1R{=}(b,j)$, $2L{=}(c,a)$, $2R{=}(i,k)$, $3L{=}(k,c)$, $3R{=}(j,b)$. Lines: $1L \to 2L$ (a), $2L \to 3L$ (c), $3L \to 2R$ (k), $2R \to 1L$ (i) – one cycle; $1R \to 3R$ (b), $3R \to 1R$ (j) – a second. $l = 2$, $h = 3$: $-$ in front of $\overline{v}_{ciak}$, which is $+$ in front of $\overline{v}_{icak}$, the form usually quoted.

Pause to think #4

(about 4 min – discuss with your neighbour)

The pairing model: the interaction only moves a pair $(p\uparrow, p\downarrow) \rightarrow (q\uparrow, q\downarrow)$, H_0 is diagonal by construction, $u = 0$.

1. Which of the fourteen diagrams can be non-zero? (Think: can the interaction connect the reference to a single? Can it scatter a particle and a hole from different pairs? What is $\tilde{f}_{pq} = U_{pq} = \sum_j \overline{v}_{pjqj}$ for an occupied and for an empty spin-orbital?)
2. Week 44 found principal -0.02300347, renormalisation -0.01258681, $E^{(3)} = -0.01041667$ at $g = 0.5$. Guess which diagrams account for which number.

Answer to pause #4

1. $\tilde{f}_{ai} = 0$ (a pair cannot be half moved) kills 1–6, 9, 13, 14; the ring needs a particle-hole scattering between different pairs and vanishes; $U_{pp} = \overline{v}_{p\bar{p}p\bar{p}} = -g/2$ for an *occupied* spin-orbital, 0 for an empty one. Left: the two ladders and the hole insertion 8.
2. Four levels, two pairs, unit spacing: doubles $i \to a$ with $D_{ia} = 2(\varepsilon_i - \varepsilon_a) = -4, -6, -2, -4$, element $-g/2$.

$$\text{pp} = -\tfrac{g^3}{8}\sum_i \Big(\sum_a \tfrac{1}{D_{ia}}\Big)^2 = -\tfrac{53}{576}g^3, \qquad \text{hh} = -\tfrac{53}{576}g^3,$$

$$\text{insertion} = 2\cdot\tfrac{g}{2}\sum_{ia}\tfrac{(g/2)^2}{D_{ia}^2} = +\tfrac{29}{288}g^3, \qquad \boxed{E^{(3)} = -\tfrac{1}{12}g^3.}$$

At $g = 0.5$: $-0.0115017 - 0.0115017 + 0.0125868 = -0.0104167$. The ladders *are* the principal term; the insertion *is* minus the renormalisation term.

Where the "correction larger than the answer" went

In the bookkeeping of week 44

The diagonal element V_{mm} of the middle vertex contains the reference energy V_{00}; the renormalisation term subtracts it. What is left, $W_{mm} = V_{mm} - V_{00}$, is the diagonal part of the ladders plus the mean-field insertions on the four lines of the double.

Pairing model

For a pair excitation the two cancel: $W_{mm} = 0$. So the hole insertion, $+g \sum_m |V_{0m}|^2/D_m^2$ with $-\tilde{f}_{ii} = g/2$ on each hole line, equals $-V_{00} \sum_m |V_{0m}|^2/D_m^2$ exactly, and the principal term is the ladders alone.

Measured

`diagrams.py`, pairing $g = 0.5$:
pp ladder -0.011501736
hh ladder -0.011501736
hole insertion $+0.012586806$
sum $-0.010416667 = E^{(3)}$

Pairing + particle-hole, $g = 0.5$, $f = 0.025$: ladders -0.0153905 each, insertion $+0.0169196$, ring 0, diagrams 2, 3, 5, 6, 9, 13, 14 switched on at $\leq 1.5 \times 10^{-4}$ ($O(f^2)$: two vertices only the particle-hole term supplies); sum $-0.0138615 = E^{(3)}$.

The renormalisation term was never a correlation correction: it removes a mean-field contribution. (The two ladders are equal because the model has as many empty levels as filled ones at the same spacing.)

The rules put to the test: a random Hamiltonian in which all fourteen are non-zero

Eight spin-orbitals, four particles, ε_p increasing, random symmetric off-diagonal u_{pq}, random Hermitian antisymmetrised $\overline{v}_{pqrs}$ (70 determinants). The fourteen diagrams are evaluated from the orbital-level elements with the rules; $E^{(3)}$ from the recursion in the full determinant space.

1 particle insertion	0.029978687	8 hole insertion	0.015920047
2 hole insertion	0.222374016	9 V closed by two crosses	0.064812430
3 two crosses closed by V	0.064812430	10 pp ladder	0.004426951
4 particle-hole scattering	0.152115743	11 hh ladder	0.002250487
5 particle line into V	−0.023773776	12 ph ring	0.008950096
6 hole line into V	−0.061323558	13 V to single via particles	−0.023773776
7 particle insertion	0.004225746	14 V to single via holes	−0.061323558
sum of the fourteen	0.399671963	$E^{(3)}$, determinant space	0.399671963

Measured

Agreement to 10^{-12}; principal −0.14176, renormalisation −0.54143, $\langle WR_0WR_0W\rangle_0 = 0.39967$. Second order: $(F,F) = -0.650749$, ring −0.146957, sum $= E^{(2)} = -0.797706$. A wrong sign or a wrong factor in any one of the fourteen would show up in the last line – which is how diagram 4 was caught during the preparation of these slides.

Reading a diagram into code: `diagrams.py`

With `eps`, `ftil`, `vbar` and the occupied/empty index lists `o`, `v`; `D1[i,a]` $= \varepsilon_i - \varepsilon_a$, `D2[i,j,a,b]` $= \varepsilon_{ij}^{ab}$:

```
v_oovv = vb[np.ix_(o, o, v, v)] # <ij|v|ab> v_vvoo = vb[np.ix_(v, v, o, o)] # <ab|v|ij>
v_vvvv = vb[np.ix_(v, v, v, v)] # <cd|v|ab> v_oooo = vb[np.ix_(o, o, o, o)] # <ij|v|kl>
v_vovo = vb[np.ix_(v, o, v, o)] # <ci|v|ak> foo, fov, fvo, fvv = blocks of ftil
# 10 particle-particle ladder: three equivalent pairs -> 1/8, h = 2, l = 2 -> +
pp = 0.125 * np.einsum("ijcd,cdab,abij,ijab,ijcd->", v_oovv, v_vvvv, v_vvoo, 1/D2, 1/D2)
# 11 hole-hole ladder
hh = 0.125 * np.einsum("klab,ijkl,abij,ijab,klab->", v_oovv, v_oooo, v_vvoo, 1/D2, 1/D2)
# 12 particle-hole ring: no equivalent lines, h = 3, l = 2 -> -
ring = -np.einsum("kjcb,ciak,abij,ijab,kjcb->", v_oovv, v_vovo, v_vvoo, 1/D2, 1/D2)
# 8 hole-line insertion on a double: one equivalent pair (a,b) -> 1/2, h = 3, l = 2 -> -
ins = -0.5 * np.einsum("kjab,ik,abij,ijab,kjab->", v_oovv, foo, v_vvoo, 1/D2, 1/D2)
# 2 cross on a hole line: h = 2, l = 1 -> -
d2 = -np.einsum("ja,ij,ai,ia,ja->", fov, foo, fvo, 1/D1, 1/D1)
```

Each line is one diagram: the matrix elements in the order top to bottom, one $1/D$ per cut, the weight and the sign in front. `GeneralHamiltonian` builds the same Hamiltonian in the determinant space and runs the recursion of week 44, so the two routes can be compared for any $(\varepsilon, u, \bar{v})$. The notebook `week45.ipynb` is self-contained and repeats every number on these frames.

Friday, lecture 4: Unlinked diagrams and the linked-diagram theorem

Fourth order: a new kind of drawing

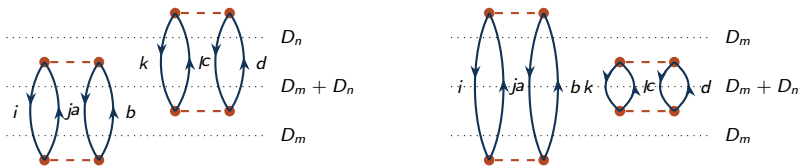
$$E^{(4)} = \langle W R_0 W R_0 W R_0 W \rangle_0 - E^{(2)} \langle W R_0^2 W \rangle_0 .$$

With four normal-ordered vertices a complete contraction can fall into two closed pieces of two vertices each:

pairing of vertices	intermediate states	
1–3 and 2–4	$m,\ m \otimes n,\ n$	unlinked, overlapping in time
1–4 and 2–3	$m,\ m \otimes n,\ m$	unlinked, overlapping in time
1–2 and 3–4	$m,\ \Phi_0,\ n$	zero: no line crosses the middle cut, Q kills it

- Each piece is a closed second-order diagram – a ring, or two crosses.
- Such a drawing is unlinked: it contains a closed part with no line connecting it to the rest.
- Everything else with four vertices is connected; the question is what the unlinked ones are worth.

The two unlinked time orderings, drawn



(a) vertices 1, 3 and 2, 4

(b) vertices 1, 4 and 2, 3

Two second-order rings, pairs (ab, ij) and (cd, kl) sharing no orbital; $D_m = \varepsilon_{ij}^{ab}$, $D_n = \varepsilon_{kl}^{cd}$ are the denominators of the rings on their own. The matrix elements, the weights and the signs of the two pieces simply *multiply* (the line and loop counts add in the exponent). Only the denominators are not local to a piece: in the middle both rings are open, and the energies *add* because the excitations involve different orbitals.

The factorisation of the denominators

Ordering (a): cuts D_m, $D_m + D_n$, D_n. Ordering (b): D_m, $D_m + D_n$, D_m. Add them:

$$\frac{1}{D_m(D_m + D_n)D_n} + \frac{1}{D_m(D_m + D_n)D_m} = \frac{1}{D_m(D_m + D_n)} \cdot \frac{D_m + D_n}{D_m D_n} = \frac{1}{D_m^2 D_n}.$$

For fixed m opened first, the two orders sum to $1/(D_m^2 D_n)$. When n opens first they give $1/(D_n^2 D_m)$. An ordered double sum already includes the interchange of m and n.

Compare with the folded term

$$E^{(2)}\langle WR_0^2 W\rangle_0 = \sum_{m,n\neq 0} \frac{W_{0n}W_{n0}}{D_n} \frac{W_{0m}W_{m0}}{D_m^2}$$

This product enters $E^{(4)}$ with a minus sign. For disjoint excitations it cancels the sum of the two overlapping orderings for each ordered pair of excitations.

For shared orbital labels, independent sums also contain formal EPV pairings. We will track their exchange partners explicitly after the proof.

Factorisation for two open excitation histories

Fix the vertex order *within* each of two independent pieces. Let A_r and B_s be their signed energy deficits after r and s vertices. Put $A_0 = B_0 = 0$. Let S_{rs} sum the denominator products of all interleavings of these prefixes, including their final cut.

$$S_{r0} = \prod_{u=1}^{r} A_u^{-1}, \quad S_{0s} = \prod_{v=1}^{s} B_v^{-1}, \quad S_{00} = 1.$$

The last vertex belongs to one of the pieces. Their deficits add on the common cut, so for $r, s > 0$,

$$S_{rs} = \frac{S_{r-1,s} + S_{r,s-1}}{A_r + B_s} = \frac{A_r + B_s}{A_r + B_s} \prod_{u=1}^{r} \frac{1}{A_u} \prod_{v=1}^{s} \frac{1}{B_v}.$$

The second equality uses the induction hypothesis. This proves factorisation for prefixes with nonzero denominators. At a closed energy graph's final vertex there is no final denominator. The preceding fourth-order example treated that endpoint explicitly.

The linked-diagram theorem for a single reference

Assume a finite orbital basis and a nondegenerate, gapped ground determinant $|\Phi_0\rangle$ of H_0. Follow its ground-state branch for small coupling λ. As before, $H(\lambda) = H_0 + \lambda(E^{(1)} + W)$ with $\langle W \rangle_0 = 0$.

Energy

For $n \geq 2$, sum the connected closed Wick diagrams of the indicated string.

$$E^{(n)} = \left[\langle W(R_0 W)^{n-1}\rangle_0\right]_{\text{connected}}, \qquad E_0(\lambda) = \varepsilon_0 + \lambda E^{(1)} + \sum_{n\geq 2} \lambda^n E^{(n)}.$$

Wave operator

With $\langle \Phi_0 | \Psi_0(\lambda) \rangle = 1$, each component of a diagram for an excited-determinant amplitude must reach an external line. Several components may survive in the same amplitude if each reaches the projecting bra.

Proof 1. Projection onto the interacting ground state

Remove the scalar shift by writing $H' = H_0 + \lambda W$. Define

$$U_T = e^{-T(H'-\varepsilon_0)}, \qquad Z(T) = \langle U_T \rangle_0 = \sum_\nu |\langle \Psi'_\nu | \Phi_0 \rangle|^2 e^{-T(E'_\nu - \varepsilon_0)}.$$

Here the spectral eigenvectors are unit normalised. The ground-state overlap is nonzero for sufficiently small λ. The gap suppresses the excited terms, giving

$$E'_0 - \varepsilon_0 = -\lim_{T\to\infty} \partial_T \log Z(T), \qquad |\Psi_0(\lambda)\rangle = \lim_{T\to\infty} \frac{U_T|\Phi_0\rangle}{Z(T)}.$$

The ratio has reference overlap one at every T. Since $\langle \Phi_0 | (H_0 - \varepsilon_0) = 0$, the energy also has a form with one marked, latest vertex:

$$-\partial_T \log Z(T) = \lambda \frac{\langle W U_T \rangle_0}{Z(T)}.$$

Proof 2. Independent vacuum components factorise

Let $W_I(t) = e^{tH_0} W e^{-tH_0}$ and let $\mathcal{T}$ put later times on the left. The Dyson series on the reference gives

$$Z(T) = \sum_{N \geq 0} \frac{(-\lambda)^N}{N!} \int_{[-T,0]^N} \mathrm{d}^N t \, \langle \mathcal{T} W_I(t_1) \cdots W_I(t_N) \rangle_0 .$$

Wick contractions partition the N labelled vertices into connected components. If there is no line between two sets, their orbital labels are summed independently and their time variables are integrated over a product domain. The signed integrand factorises because every closed component has even fermion parity.

Relative time orderings

For two components the Cartesian product of their ordered time domains splits into all interleavings that preserve each internal ordering. One ordering on its own need not factorise. Integrating over their union gives the product of the two component integrals.

Proof 3. Counting the components produces an exponential

Group connected vacuum graphs by type α. Write $C_\alpha(T)$ for the signed, integrated weight of one component, with its vertex factorial and internal symmetry factors. Suppose m_α components of type α occur, each with r_α vertices, so $N = \sum_\alpha m_\alpha r_\alpha$. The number of partitions of the vertex labels is

$$\frac{N!}{\prod_\alpha (r_\alpha!)^{m_\alpha} m_\alpha!}.$$

The Dyson $1/N!$ cancels the numerator. Each $1/r_\alpha!$ belongs to one component. Permuting identical components leaves the factor $1/m_\alpha!$. Summing their multiplicities gives

$$Z(T) = \prod_\alpha \sum_{m \geq 0} \frac{C_\alpha(T)^m}{m!} = \exp\Big[\sum_\alpha C_\alpha(T)\Big].$$

For example, $\langle W \rangle_0 = 0$ implies $Z_1 = 0$. If $Z = 1 + \sum_n \lambda^n Z_n$, the fourth-order coefficient of $\log Z$ is $Z_4 - Z_2^2/2$. The identical pair of second-order vacuum components cancels exactly.

Proof 4. The energy selects one connected component

In $\langle W U_T \rangle_0$, distinguish the vertex W at time zero. Every fully contracted term has a unique component containing this vertex. All other components are ordinary vacuum components. For N integrated vertices, choosing the k that join the marked component gives

$$\frac{1}{N!}\binom{N}{k} = \frac{1}{k!(N-k)!}.$$

The unmarked vertices therefore generate exactly $Z(T)$. If $M(T)$ sums the components attached to the marked vertex, then

$$\langle W U_T \rangle_0 = M(T) Z(T), \qquad E_0' - \varepsilon_0 = \lambda \lim_{T \to \infty} M(T).$$

Every interaction in M is joined to the marked vertex by a path of fermion lines and interactions. Closing the external state with the reference gives a connected vacuum diagram. Equivalently, $-\partial_T \log Z$ selects the same connected series.

Proof 5. Imaginary-time intervals give the Goldstone denominators

Order the $n-1$ integrated vertices beneath the marked vertex $t_n = 0$. During $s_k = t_{k+1} - t_k$ the free propagation supplies $e^{-\Delta_k s_k}$, with

$$\Delta_k = \sum_{a \text{ crossing}} \varepsilon_a - \sum_{i \text{ crossing}} \varepsilon_i = -D_k > 0.$$

Every cut of a connected closed graph crosses a line. At finite T, $s_k \geq 0$ and $\sum_k s_k \leq T$. As $T \to \infty$ the ordered integral, including its Dyson signs, becomes

$$(-1)^{n-1} \int_{[0,\infty)^{n-1}} \prod_{k=1}^{n-1} e^{-\Delta_k s_k} \mathrm{d}s_k = \prod_{k=1}^{n-1} \frac{-1}{\Delta_k} = \prod_{k=1}^{n-1} \frac{1}{D_k}.$$

Thus each connected, time-ordered graph has exactly the denominators, Wick sign and weight derived on Thursday. Adding $\lambda E^{(1)}$ restores the energy of $H(\lambda)$. The small-coupling ground-state expansion is unique, so these are the RS energy coefficients.

Proof 6. Vacuum components cancel in the wave operator

Project onto an excited determinant and divide by the vacuum amplitude:

$$c_\mu(\lambda) = \lim_{T\to\infty} \frac{\langle \Phi_\mu | U_T | \Phi_0 \rangle}{Z(T)}.$$

Represent the bra by its external annihilators and creators at time zero. In every numerator graph, separate the components that reach at least one external line from those that reach none. Choosing the vertices in the second set produces the same binomial factor as in the energy proof. Their sum is $Z(T)$ and cancels at finite T.

$$|\Psi_0(\lambda)\rangle = |\Phi_0\rangle + \sum_{n\geq 1} \lambda^n \big[(R_0 W)^n |\Phi_0\rangle \big]_{\mathrm{linked}}.$$

Here brackets instruct us to retain the indicated Wick diagrams with their line denominators. The open graph has one more interval, between its last interaction and its external bra. This supplies the leftmost R_0. Distinct open components may end on different external legs.

A disconnected contribution to a linked amplitude

Let m and n be independent double excitations on disjoint orbitals with deficits D_m and D_n. Project onto their combined quadruple excitation and fix the same determinant phase in both time orderings. Even excitation operators on disjoint orbitals commute. The two terms have denominators

$$\frac{1}{D_m(D_m + D_n)} + \frac{1}{D_n(D_m + D_n)} = \frac{1}{D_m D_n}.$$

Multiplying by the two creation matrix elements gives the product of the first-order doubles amplitudes (with the chosen determinant phase). Each component reaches the quadruple bra. A detached vacuum bubble would instead multiply the numerator by a factor already present in $Z(T)$.

Further reading: J. Goldstone, Proc. R. Soc. A **239** (1957). I. Shavitt and R. J. Bartlett, *Many-Body Methods in Chemistry and Physics*, §5.7.5 (EPV terms) and chapter 6 (linked diagrams).

Pauli exclusion at an intermediate cut

Consider two overlapping single excitations $X = a_a^\dagger a_i$ and $Y = a_b^\dagger a_i$, with $a \neq b$. Each excitation is allowed on its own. Together they would empty the same occupied spin orbital twice:

$$YX|\Phi_0\rangle = a_b^\dagger a_i a_a^\dagger a_i|\Phi_0\rangle = -a_b^\dagger a_a^\dagger a_i^2|\Phi_0\rangle = 0.$$

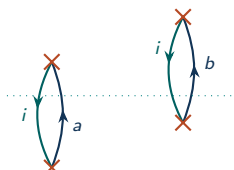
Creating two particles in the same virtual spin orbital also gives zero, since $(a_a^\dagger)^2 = 0$. Reusing a label *after* its line has ended can describe a legitimate excitation.

Where an exclusion-principle-violating (EPV) term enters

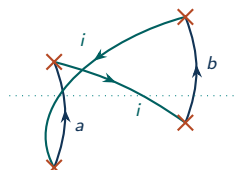
A determinant sum contains only admissible intermediate states. An individual Wick pairing sums the orbital labels of its lines independently. Two distinct lines present at the same cut can then acquire the same hole or particle label. The cut in that individual term does not represent a determinant.

Direct and exchanged hole lines at fourth order

Four one-body vertices create a, b in turn and remove them in the same order. Both hole lines (teal) carry i across the dotted cut. Crosses mark the vertices.



direct: $l = 2$



exchange: $l = 1$

$$D_a = \varepsilon_i - \varepsilon_a, \quad D_b = \varepsilon_i - \varepsilon_b, \qquad F = \frac{|\tilde{f}_{ai}\tilde{f}_{bi}|^2}{D_a(D_a + D_b)D_b}.$$

Exchanging the upper hole endpoints leaves the numerator and cut energies unchanged. With $h = 2$, the signs give $+F - F = 0$, as required by $a_i^2 = 0$. The middle deficit $D_a + D_b$ is a formal line sum.

What happens to the exchange term after regrouping?

Fix a shared-hole assignment and include the relative time orderings needed for factorisation. Write U_{P} for the unlinked sum with Pauli-allowed overlaps, and U_{EPV} for its formal shared-label extension.

$$\underbrace{(U_{\mathrm{P}} + U_{\mathrm{EPV}})}_{\text{all unlinked pairings}} + \underbrace{L_{\mathrm{EPV}}}_{\text{linked exchange partners}} - \underbrace{(U_{\mathrm{P}} + U_{\mathrm{EPV}})}_{\text{folded subtraction}} = L_{\mathrm{EPV}}.$$

For the pair of contractions just drawn, $L_{\mathrm{EPV}} = -U_{\mathrm{EPV}}$ after summing matching orders. In the principal term alone, the forbidden occupation has zero net weight. Once the unlinked diagrams cancel against the folded subtraction, the linked exchange contribution remains in the energy.

How to perform the sums

Use Pauli-allowed determinants in the original resolvent formula. In the connected diagram formula, sum each line over its full orbital space. A separate veto on coincident labels across different vertices would erase the exchange contribution. Antisymmetry already gives $\overline{v}_{aaij} = \overline{v}_{abii} = 0$ at a single vertex.

The two-level benchmark separates the two calculations

Take one fermion in orbitals i, a with $\Delta = \varepsilon_a - \varepsilon_i > 0$, $D = -\Delta$, and off-diagonal mixing g. Exact diagonalisation gives

$$H = \begin{pmatrix} \varepsilon_i & g^* \\ g & \varepsilon_a \end{pmatrix}, \qquad E_- - \varepsilon_i = \frac{\Delta - \sqrt{\Delta^2 + 4|g|^2}}{2} = -\frac{|g|^2}{\Delta} + \frac{|g|^4}{\Delta^3} + O(|g|^6).$$

- **Determinants.** $QWQ = 0$, hence $\langle WR_0WR_0WR_0W\rangle_0 = 0$. The folded term gives the entire fourth-order correction $-E^{(2)}\langle WR_0^2W\rangle_0 = -|g|^4/D^3 > 0$.
- **Wick contractions.** Two simultaneous excursions cannot form a physical state in this one-fermion model. In the unrestricted expansion, the direct and exchange pairings cancel in the principal term. Removing the unlinked sum against the folded term leaves the linked exchange sum with the same coefficient $+|g|^4/\Delta^3$.

Counting the two-level diagrams gives the exact coefficient

For a single virtual orbital, the overlapping ordering creates the same excitation at vertices 1 and 2 and removes it at 3 and 4. The three cut deficits are $D, 2D, D$. Before its Wick sign, each pairing has value $F = |g|^4/(2D^3)$.

Matching of particles and holes	pairings	loops	signed sum
Both use the same matching	2	2	$+2F$ (unlinked)
They use opposite matchings	2	1	$-2F$ (linked)

Each species has two possible matchings between the lower and upper vertices. All four have $h = 2$. The principal term gives $2F - 2F = 0$. The folded subtraction cancels $+2F$, leaving

$$E^{(4)} = -2F = -\frac{|g|^4}{D^3} = \frac{|g|^4}{\Delta^3}.$$

Deleting the EPV assignments in the linked sum would incorrectly give zero.

A numerical check: two non-interacting copies

Measured

Two non-interacting copies A, B of a three-level pairing system with one pair each, $g = 0.5$ (no shared orbitals, no EPV terms). Restricting the intermediate states of the principal fourth-order term to (A, AB, B), (A, AB, A), (B, AB, A), (B, AB, B): -0.00183105. $2E_A^{(2)}\langle WR_0^2W\rangle_A$: -0.00183105. $E_{AB}^{(4)} = 0.00109863 = 2E_A^{(4)}$ to fourteen digits. (`diagrams.py`, `fourth_order_unlinked`; the (A, AB, B) and (A, AB, A) orderings alone are -0.000437 and -0.000478 – neither is a product, only their sum is.)

- With disjoint orbitals the unperturbed energies add, $D_{m\otimes n} = D_m + D_n$, and matrix elements and signs factorise; the (A, AB, B) and (A, AB, A) patterns have the denominators of the factorisation identity.
- Their sum is $\langle WR_0^2W\rangle_A\, E_B^{(2)}$; the other two give $E_A^{(2)}\langle WR_0^2W\rangle_B$. The folded term of the two copies, $(E_A^{(2)} + E_B^{(2)})(\langle WR_0^2W\rangle_A + \langle WR_0^2W\rangle_B)$, has exactly these cross terms: the cross-copy part of $E^{(4)}$ cancels and $E_{AB}^{(4)} = E_A^{(4)} + E_B^{(4)}$.

Pause to think #5

(about 3 min – discuss with your neighbour)

Brillouin-Wigner theory uses $R_E = Q/(E_0 - H_0)$: the *exact* energy in every denominator, and no renormalisation terms at all.

1. Redo the factorisation identity with $D_m \to D_m + \Delta E$, $D_n \to D_n + \Delta E$, $D_m + D_n \to D_m + D_n + \Delta E$. Does it still factorise?
2. What does this say about the unlinked diagrams of a truncated BW calculation, and about its scaling with system size?

Answer to pause #5

1. No. With $d_m = D_m + \Delta E$ etc. the middle denominator is $D_m + D_n + \Delta E \neq d_m + d_n$ (it is $d_m + d_n - \Delta E$), so $\frac{1}{d_m(d_m+d_n-\Delta E)}\left(\frac{1}{d_n} + \frac{1}{d_m}\right) \neq \frac{1}{d_m^2 d_n}$. The excursion deficits add for disjoint excitations in RS. BW inserts the same global energy shift once in each intermediate-state denominator.
2. The cancellation used in the RS proof fails in general for self-consistent BW truncated at finite order. Spurious cross terms can spoil additivity for two independent copies, as in the week-44 example. The exact energies of two non-interacting copies are additive.

Consequences for independent systems

- **Additivity at each order.** For $H = H_A + H_B$ on disjoint non-interacting fragments and a product reference, $Z_{AB} = Z_A Z_B$. Thus $\log Z_{AB} = \log Z_A + \log Z_B$ and $E_{AB}^{(n)} = E_A^{(n)} + E_B^{(n)}$. For M identical independent copies the coefficient is $ME_A^{(n)}$.
- **Truncated BW.** The global energy in the denominators spoils the RS factorisation identity in general. A finite self-consistent BW truncation can lose size extensivity.
- **Summation by topology.** Ladders and rings are families of connected energy graphs. In coupled cluster, products of excitations in $e^T|\Phi_0\rangle$ supply the disconnected open pieces that survive in the wave function.

The theorem is an order-by-order identity. Convergence of the energy series at a particular coupling is a separate question.

Summary

Summary of week 45

Thursday: the rules, derived

- The complete lists: $\Omega^{(1,2,3)}$, $E^{(1,2,3,4)}$, renormalisation terms as products of pieces
- $H_I = E^{(1)} + W$: the recursion in W has no $E^{(1)}$ terms; $E^{(3)} = \langle WR_0WR_0W\rangle_0$, $E^{(4)} = \langle WR_0WR_0WR_0W\rangle_0 - E^{(2)}\langle WR_0^2W\rangle_0$
- Contractions are lines (two non-zero contractions, arrows); cuts are denominators ($[H_0, a_p^\dagger] = \varepsilon_p a_p^\dagger$); Q kills time-separated pieces only
- Weight: 4^{n_V} relabellings against $(\frac{1}{4})^{n_V}$, equivalent lines halve the count. Sign: ends, π, cycles = loops, $(-1)^{K-l}(-1)^p = (-1)^{h+l}$

Friday: third order and the theorem

- Fourteen connected third-order diagrams; MP3 = pp ladder + hh ladder + ph ring; pairing model $-\frac{53}{576} - \frac{53}{576} + \frac{29}{288} = -\frac{1}{12}$ (g^3); all fourteen checked to 10^{-12}
- Unlinked fourth-order diagrams; $\frac{1}{D_m(D_m+D_n)D_n} + \frac{1}{D_m(D_m+D_n)D_m} = \frac{1}{D_m^2 D_n}$; the folded term is the unlinked part with a minus sign
- Theorem: ΔE = connected closed diagrams, Ψ_0 = linked diagrams. Extensivity; BW does not factorise; sum by topology (ladders, rings, CC)

Vacuum normalisation cancels detached closed pieces. Linked exchange terms with coincident labels remain in the unrestricted sums.

Next week

Week 46, November 12–13: further developments

Week 45 established the energy and wave-operator forms of the theorem by vacuum normalisation. Week 46 can revisit the time-independent recursion and the adiabatic Gell-Mann–Low construction, then use ladders and rings to motivate the coupled-cluster expansion of chapter 10.

Reading: chapter 9, section 9.15; Shavitt and Bartlett, chapters 4–5 (and 6 for the time-dependent proof). The week-45 exercises (last frames, `Exercises/ExercisesWeek45.tex`) are discussed in the exercise session and answered on the week-46 slides.

Answers to the week-44 exercises

Exercise 1: the energy as a projection, and a)

$H = H_0 + V$, $H_0|\phi_n\rangle = \varepsilon_n|\phi_n\rangle$, $H|\psi_n\rangle = E_n|\psi_n\rangle$; ground states non-degenerate and $\langle\phi_0|\psi_0\rangle \neq 0$. Nothing is assumed about the norm of $|\psi_0\rangle$.

The projection

Project $H|\psi_0\rangle = E_0|\psi_0\rangle$ onto $\langle\phi_0|$; $\langle\phi_0|H_0 = \varepsilon_0\langle\phi_0|$ (Hermiticity), so

$$(E_0 - \varepsilon_0)\langle\phi_0|\psi_0\rangle = \langle\phi_0|V|\psi_0\rangle \quad \Longrightarrow \quad \boxed{E_0 - \varepsilon_0 = \frac{\langle\phi_0|V|\psi_0\rangle}{\langle\phi_0|\psi_0\rangle}.}$$

Exact; a transition element, not an expectation value, hence not variational. Non-degeneracy is not needed here – it is needed when $z - H_0$ is inverted on the Q space.

a) Idempotency

$P^2 = |\phi_0\rangle\langle\phi_0|\phi_0\rangle\langle\phi_0| = P$; $Q^2 = (1-P)^2 = 1 - 2P + P^2 = 1 - P = Q$. Also Hermitian, $PQ = QP = 0$, $P + Q = 1$, and $[P, H_0] = [Q, H_0] = 0$ because $|\phi_0\rangle$ is an eigenstate of H_0: $Q = \sum_{n\neq 0} |\phi_n\rangle\langle\phi_n|$, and any $f(H_0)$ is diagonal in the same basis, so $Qf(H_0)Q = Qf(H_0) = f(H_0)Q$ – the licence to write $Q/(z - H_0)$ (week 44, lecture 1).

Exercise 1 b): the expansion for arbitrary z

$|\psi_0\rangle = (P+Q)|\psi_0\rangle = \langle\phi_0|\psi_0\rangle|\phi_0\rangle + Q|\psi_0\rangle$. Add and subtract $z|\psi_0\rangle$ in the Schrödinger equation:

$$(z - H_0)|\psi_0\rangle = (z - E_0 + V)|\psi_0\rangle.$$

Multiply by Q; on the Q space $z - H_0$ has eigenvalues $z - \varepsilon_n \neq 0$ ($n \neq 0$) and can be inverted:

$$\frac{Q}{z-H_0} \equiv Q(z-H_0)^{-1}Q = \sum_{n\neq 0} \frac{|\phi_n\rangle\langle\phi_n|}{z-\varepsilon_n}, \qquad Q|\psi_0\rangle = \frac{Q}{z-H_0}(z-E_0+V)|\psi_0\rangle.$$

(At $z = \varepsilon_0$ the full inverse has a pole; the reduced one exists iff the ground state of H_0 is non-degenerate – *this* is where the assumption enters.) Call $A(z) = \frac{Q}{z-H_0}(z - E_0 + V)$ and iterate:

$$|\psi_0\rangle = \langle\phi_0|\psi_0\rangle\big(1 + A + \cdots + A^N\big)|\phi_0\rangle + A^{N+1}|\psi_0\rangle \xrightarrow{N\to\infty} \boxed{|\psi_0\rangle = \langle\phi_0|\psi_0\rangle \sum_{n=0}^{\infty}\Big(\frac{Q}{z-H_0}(z-E_0+V)\Big)^n|\phi_0\rangle,}$$

a formal Neumann series (convergence not guaranteed); the overlap is an overall factor, as linearity demands. Insert into the projected identity:

$$\boxed{E_0 = \varepsilon_0 + \sum_{n=0}^{\infty}\langle\phi_0|V\Big(\frac{Q}{z-H_0}(z-E_0+V)\Big)^n|\phi_0\rangle.}$$

The overlap has cancelled – which is why one may set $\langle\phi_0|\psi_0\rangle = 1$ from the start. Still implicit: E_0 sits inside $z - E_0 + V$ for every z but one.

Exercise 1 c): the two schemes

$z = E_0$: Brillouin-Wigner

$z - E_0 + V = V$: only interactions and resolvents alternate,

$$E_0 = \varepsilon_0 + V_{00} + \sum_{m\neq 0} \frac{V_{0m}V_{m0}}{E_0 - \varepsilon_m} + \sum_{m,n\neq 0} \frac{V_{0m}V_{mn}V_{n0}}{(E_0 - \varepsilon_m)(E_0 - \varepsilon_n)} + \cdots$$

Explicit and simple, but E_0 in every denominator: implicit, self-consistent; a truncation after k vertices is *not* a polynomial of degree k in the coupling. The full resummation (Feshbach partitioning, week 44) is exact. Finite truncations are not size extensive.

$z = \varepsilon_0$: Rayleigh-Schrödinger

$R_0 = Q/(\varepsilon_0 - H_0)$ is known, but $z - E_0 + V = V - \Delta E$ and ΔE must be expanded too ($R_0 \Delta E|\phi_0\rangle = 0$ removes the rightmost one); with $D_m = \varepsilon_0 - \varepsilon_m$,

$$E^{(1)} = V_{00}, \qquad E^{(2)} = \sum_{m\neq 0} \frac{V_{0m}V_{m0}}{D_m}, \qquad E^{(3)} = \sum_{m,n\neq 0} \frac{V_{0m}V_{mn}V_{n0}}{D_m D_n} - V_{00}\sum_{m\neq 0} \frac{V_{0m}V_{m0}}{D_m^2}.$$

The first two orders are the BW terms with $E_0 \to \varepsilon_0$; the third carries the *renormalisation* term, which has no BW counterpart.

Exercise 1 c): comparing the first few terms

Expand a Brillouin-Wigner denominator around the unperturbed one,

$$\frac{1}{E_0-\varepsilon_m}=\frac{1}{(\varepsilon_0-\varepsilon_m)+\Delta E}=\frac{1}{\varepsilon_0-\varepsilon_m}-\frac{\Delta E}{(\varepsilon_0-\varepsilon_m)^2}+\cdots, \qquad \Delta E\simeq V_{00} \text{ to this order,}$$

and insert it into the second-order BW term:

$$\sum_{m\neq 0}\frac{V_{0m}V_{m0}}{E_0-\varepsilon_m}\simeq\sum_{m\neq 0}\frac{V_{0m}V_{m0}}{\varepsilon_0-\varepsilon_m}-V_{00}\sum_{m\neq 0}\frac{V_{0m}V_{m0}}{(\varepsilon_0-\varepsilon_m)^2}=E^{(2)}_{\mathrm{RS}}+\left[\text{renormalisation term of } E^{(3)}_{\mathrm{RS}}\right].$$

- Through third order the two schemes agree term by term once the BW denominators are expanded: the RS renormalisation terms *are* the expanded energy dependence of the BW denominators.
- At a *finite* order the difference is real: RS is a true polynomial in the coupling and size extensive (this week: the renormalisation terms cancel the unlinked diagrams); truncated BW contains partial resummations to all orders and is not.
- In a small problem BW often converges faster – for the two-level model with purely off-diagonal coupling the two-vertex BW equation $E_0-\varepsilon_1=V_{12}^2/(E_0-\varepsilon_2)$ is already the exact secular equation while RS needs all even orders – but for many-body systems the wrong scaling with particle number is fatal, and RS (the linked-diagram expansion) is the one that is used.

Exercise 2 a): two particles in a single j-shell, Brillouin-Wigner

Pair orbitals $|m,-m\rangle$, $m=\frac{1}{2},\ldots,j$: $\Omega = j+\frac{1}{2} \geq 2$ pair states, all degenerate in H_0 ($2\varepsilon_j \equiv 0$); $\langle m,-m|v|m',-m'\rangle = -G$ for *all* m,m'. In this basis

$$H = -G\,J, \qquad J = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & & \vdots \\ 1 & \cdots & 1 \end{pmatrix}_{\Omega\times\Omega}.$$

Reference $|\phi_0\rangle = |m_0,-m_0\rangle$, $\varepsilon_0 = 0$; $V_{00} = V_{0m} = V_{mn} = -G$; every BW denominator is $E_0 - 0 = E_0 \equiv x$. The term with k vertices has k factors $-G$ and $k-1$ free sums over the $\Omega-1$ states of Q:

$$x = -G\sum_{k=1}^{\infty}\Big[\frac{-(\Omega-1)G}{x}\Big]^{k-1} = \frac{-Gx}{x+(\Omega-1)G} \implies x\,(x+\Omega G) = 0 \implies \boxed{E_0 = -\Omega G = -(j+\tfrac{1}{2})G,}$$

and at this root $(\Omega-1)G/|x| = (\Omega-1)/\Omega < 1$: the geometric series converges. The resummed form of week 44 gives it at once: $v_0 = (V_{m0})$ is an eigenvector of $J_{\Omega-1}$ with eigenvalue $\Omega-1$, so $E_0 = -G + (\Omega-1)G^2/(E_0+(\Omega-1)G)$, i.e. $E_0(E_0+\Omega G) = 0$.

A trap to avoid

Naive fixed-point iteration $E \leftarrow f(E) = -G + (\Omega-1)G^2/(E+(\Omega-1)G)$ converges to the *unphysical* root 0: $|f'(-\Omega G)| = \Omega - 1 > 1$ (repelling), $|f'(0)| = 1/(\Omega-1) < 1$ (attracting). Solve the implicit equation as an equation (bisection, Newton, or multiply out).

Exercise 2 b): direct diagonalisation

$J = \Omega\,|s\rangle\langle s|$ with $|s\rangle = \Omega^{-1/2}(1,\ldots,1)$ has rank one: eigenvalues Ω (once) and 0 ($\Omega - 1$ times). Hence

$$E_0 = -\Omega G = -(j+\tfrac{1}{2})G \text{ (once)}, \qquad E = 0 \ (\Omega - 1 \text{ times}), \qquad |\psi_0\rangle = \frac{1}{\sqrt{\Omega}} \sum_{m>0} |m,-m\rangle.$$

- The Brillouin-Wigner result of a) is the exact ground-state energy; the other BW root, 0, is also an eigenvalue.
- The ground state is the coherent superposition of all pair states – the *collective pair* of the seniority model. It gains the full $-\Omega G$ because the constant attractive interaction is maximally coherent in this combination; the $\Omega - 1$ orthogonal combinations (seniority two) feel no interaction at all.
- For readers who know the quasispin formalism: the seniority-zero energy for N particles in a degenerate shell is $-\frac{G}{4}N(2\Omega - N + 2)$, which at $N = 2$ gives $-G\Omega$.

Measured

$j = \frac{5}{2}, \frac{7}{2}, \frac{9}{2}$ ($\Omega = 3, 4, 5$), $G = 1$: eigenvalues $-3, -4, -5$; the two BW roots by bisection $-\Omega$ and 0 to twelve digits; plain fixed-point iteration from $-\Omega - 0.37$ ends at 0 every time (`week45.ipynb`; also `Exercises/AnswersWeek44.tex`).

Exercise 2 b): why Rayleigh-Schrödinger cannot even start, and the remedy

Rayleigh-Schrödinger

$$E^{(2)} = \sum_{m\neq 0} \frac{V_{0m}V_{m0}}{\varepsilon_0 - \varepsilon_m} = \sum_{m\neq 0} \frac{G^2}{0} : \text{ undefined.}$$

The reduced resolvent $Q/(\varepsilon_0 - H_0)$ does not exist: the reference is not isolated in the spectrum of H_0, which is exactly the assumption exercise 1 was built on (week 44, lecture 1, step 4). Brillouin-Wigner survives because its denominators hold the *exact* $E_0 - \varepsilon_m = -\Omega G \neq 0$: the interaction itself lifts the degeneracy, and the self-consistent equation knows it.

The remedy: degenerate perturbation theory

Enlarge the model space to the whole multiplet, $P = \sum_m |m,-m\rangle\langle m,-m|$, and diagonalise V inside it before treating anything perturbatively. Here the multiplet is the entire space, so that diagonalisation *is* the exact solution and nothing is left to expand. With several non-degenerate shells the same idea gives the multi-reference (model-space) theories and the effective interaction $H_{\text{eff}} = PHP + PHQ(E - QHQ)^{-1}QHP$ of week 44 in a model space of several determinants – the $\hat{Q}$-box of lecture 1, step 4.

Exercises, week 45

Exercises week 45 (1): the expansion, and the Hamiltonian in normal-ordered form

Let $\hat{H} = \hat{H}_0 + \hat{H}_I$, $|\Phi_n\rangle$ the eigenstates of $\hat{H}_0$ and $|\Psi_n\rangle$ those of $\hat{H}$; ground states non-degenerate, $E_0 - \varepsilon_0 = \langle\Phi_0|\hat{H}_I|\Psi_0\rangle/\langle\Phi_0|\Psi_0\rangle$, $\hat{P} = |\Phi_0\rangle\langle\Phi_0|$, $\hat{Q} = 1 - \hat{P}$.

a) Show that for any ω

$$E_0 = \varepsilon_0 + \sum_{n=0}^{\infty}\langle\Phi_0|\hat{H}_I\Big(\frac{\hat{Q}}{\omega - \hat{H}_0}(\omega - E_0 + \hat{H}_I)\Big)^n|\Phi_0\rangle.$$

b) Discuss the result for $\omega = E_0$ (Brillouin-Wigner) and $\omega = \varepsilon_0$ (Rayleigh-Schrödinger); compare the first few terms and discuss the differences.

c) Show that $\hat{H}_0 = \sum_{pq}\langle p|\hat{h}_0|q\rangle a_p^\dagger a_q = \sum_{pq}\langle p|\hat{h}_0|q\rangle\{a_p^\dagger a_q\} + \sum_i\langle i|\hat{h}_0|i\rangle$, and that

$$\hat{H}_I = \frac{1}{4}\sum_{pqrs}\langle pq|\hat{v}|rs\rangle a_p^\dagger a_q^\dagger a_s a_r = \frac{1}{4}\sum_{pqrs}\langle pq|\hat{v}|rs\rangle\{a_p^\dagger a_q^\dagger a_s a_r\} + \sum_{pqi}\langle pi|\hat{v}|qi\rangle\{a_p^\dagger a_q\} + \frac{1}{2}\sum_{ij}\langle ij|\hat{v}|ij\rangle.$$

Explain the meaning of the various symbols. Which reference vacuum has been used? Write down the diagrammatic representation of all these terms.

d) Use the diagrammatic representation of the Hamiltonian to set up all diagrams (antisymmetrised Goldstone or Hugenholtz) to second order, including the reference energy, that contribute to E_0 in Rayleigh-Schrödinger theory. Use the diagram rules to write down their closed-form expressions.

(`Exercises/ExercisesWeek45.tex`; a) and b) revisit exercise 1 of week 44 – compare your answer with the one given above.)

Exercises week 45 (2): a one-particle system drawn

Consider a one-particle system with $\hat{H} = \hat{H}_0 + \hat{H}_I$,

$$\hat{H}_0 = \sum_p \varepsilon_p a_p^\dagger a_p, \qquad \hat{H}_I = g \sum_{pq} a_p^\dagger a_q, \qquad \langle p|\hat{h}_0|q\rangle = \delta_{pq}\varepsilon_p,$$

g real, two one-particle states with $\varepsilon_1 < \varepsilon_2$; $p = 1$ is the model space, $p = 2$ the excluded space. Hole labels $ijk\ldots$ (below the Fermi level), particle labels $abc\ldots$ (above).

e) Use c) to write the Hamiltonian in normal-ordered form and set up all diagrams. Use an X for the interaction $\hat{H}_I$.

f) With the ground state (model space) $|\Phi_0\rangle = a_i^\dagger|0\rangle = a_1^\dagger|0\rangle$ and the excited state $|\Phi_i^a\rangle = a_a^\dagger a_i|\Phi_0\rangle$, $a = 2$, $i = 1$: set up the 2×2 Hamiltonian matrix, find the exact energy and expand the ground-state result in powers of g.

g) Find the ground-state energy to third order in Rayleigh-Schrödinger perturbation theory and compare with the expansion of the exact energy. Write down all diagrams which contribute and comment on your results.

Many-body connection

This is the two-state model of week 44 ($E_- = \varepsilon_1 + g - g^2/\Delta + 0 \cdot g^3 + g^4/\Delta^3 + \cdots$); the new part is to *draw* it: a cross X carrying g on a line, a reference-energy bubble, and the third-order cancellation between the principal term and the renormalisation term seen as the scalar part of the middle vertex (Thursday, step zero). Answers on the week-46 slides.